COURSE 2 · SECTION 12 · EXERCISE

Rework, late gates and the valve program

Section 12 · Design program status dashboard (portfolio project 1)

Start: Engxtech_S12.pbix (your finished S12 file); save a copy as Engxtech_S12_Exercise.pbix

Time: about 25 min

Data: the course data folder

DESIGN QUESTION

How many gates needed rework or finished late, where does the motorised control valve's program slip, and how did the change backlog move between the end of 2025 and mid-2026?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

In video S12 you built the Program status page from a blank file: NPD_Stages with Plan Days, Actual Days, Slip Days and On Time; Drawings with DaysLate; the cleaned Changes table with CycleDays; a marked Dates table; and eleven measures in _Measures, including Open Backlog (as-of, counted at month end). This exercise adds two gate measures and uses the slicers to answer follow-up questions.

New in this exercise (not shown in the video; one line each):

- **Gate Days Late**: a column: DATEDIFF from PlannedEnd to ActualEnd; positive = the gate finished late
- **DISTINCTCOUNT**: counts the different values of a column in the current filter, here the revisions with a rework gate

Your tasks

1. In your exercise copy, add the measures **Rework Gates** (gates with GateResult = Rework) and **Revs With Rework** (distinct PartRev among them).
2. Add the column **Gate Days Late** to NPD_Stages and the measure **Late Gates** (gates with Gate Days Late above 0). Which single gate finished latest?
3. Product slicer = Motorised control valve: read On-time Gate %, the stage with the most slip, and the open ECRs.
4. Clear the slicer. Hover the backlog line at Dec 2025 and Jun 2026.
5. Part slicer = P300: how many changes are closed, and what are their median and p90 cycle days?
6. Write one sentence for the weekly review that uses two of these numbers.

Check your answer

Check	Expected
Rework gates / revisions with rework	**16 / 8**
Late gates	**84**
Latest gate	**P300-B / Production handover / 22 d**
Motorised control valve: On-time Gate %	**62.1%**
Motorised control valve: top slip stage	**Checking, 31 d**
Motorised control valve: open ECRs	**15**
Open Backlog, Dec 2025 / Jun 2026	**15 / 26**
P300: closed / median / p90 cycle days	**26 / 47.5 / 62.5**

Hint. Late Gates counts gates that ended after plan; Slip Days counts days a stage ran over its own plan. A stage can start late, end late and still show 0 slip. If the product slicer changes nothing, check that the Parts-BOM relationship is set to Cross filter Both.

Stuck? The worked solution with the full DAX and M is in Solutions/S12_Solution.pdf.